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Birth Quarter, Playing Position, and Market Value in Elite International Football: Evidence from the FIFA World Cup 2026

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Abstract

The relative age effect is well documented in football, but unequal birth-quarter representation does not necessarily translate into positional or economic differences among elite players. This study examined associations between birth quarter, playing position, and market value among 1248 players from the 48 national teams participating in the 2026 FIFA World Cup. Publicly available FIFA and Transfermarkt data, with market valuations retrieved on 10 August 2026, were analysed using chi-square goodness-of-fit tests, multivariable logistic regression, and linear models adjusted for age, height, continental confederation, and, for market value, age^2 and playing position. Economic analyses included 827 players and incorporated a national-team random intercept. Players born early in the year were overrepresented, but no birth-quarter comparisons in the primary adjusted models reached statistical significance for playing position or market value. For goalkeepers, the comparison between the fourth (Q4) and first (Q1) birth quarters yielded an odds ratio of 0.53 (95% confidence interval: 0.28–1.01). Adjusted market-value differences relative to Q1 were +15.9% for the second birth quarter (Q2), −1.4% for the third (Q3), and +10.7% for Q4. National-team clustering accounted for 35.2% of the residual variance in log-transformed market value not explained by the fixed effects. These findings distinguish unequal representation from adjusted positional and economic associations within an already selected elite sample. However, confidence intervals included potentially relevant associations, and the cross-sectional design cannot establish how relative age influences earlier selection and progression.

Keywords: relative age effect; birth quarter; elite football; playing position; market value; FIFA World Cup; multilevel modelling

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1. Introduction

The relative age effect (RAE) describes the overrepresentation of athletes born early in the selection year and the corresponding underrepresentation of those born towards the end of the same cohort. In football, this phenomenon has been widely documented during youth development and remains observable, although generally to a lesser extent, in professional football. Its origins have been linked to the interaction between temporary

differences in growth and maturation, participation opportunities, and talent identification and selection processes that may initially favour relatively older players. However, the persistence of an uneven birthdate distribution at the elite level does not, in itself, establish whether relative age remains associated with relevant professional career characteristics after the main development and selection processes have been completed [1–4].

The distinction between being selected more frequently and subsequently achieving better outcomes is central to recent RAE research. Longitudinal evidence shows that development pathways towards elite football are considerably more complex than a linear accumulation of advantages. Analysing 9527 players from European men's national teams, Brustio et al. [5] showed that only a limited proportion of players selected at youth levels subsequently reached the senior national team. Similarly, recent studies have found that relatively younger players who progress through selection processes may exhibit transition probabilities or career outcomes comparable to, or even better than, those of their relatively older peers [6–8]. These findings suggest that RAE-related overrepresentation and its potential consequences in adulthood should be regarded as related but conceptually distinct dimensions.

One potential consequence of early relative-age advantages concerns positional specialisation. During youth development, temporary differences in maturation and anthropometry may influence both selection and allocation to specific playing roles. If early positional assignments shape subsequent role-specific training and competitive experience, their consequences could persist into senior football even after the initial maturational differences have diminished. Birth-quarter differences in senior positional representation might therefore reflect accumulated developmental opportunities rather than a continuing physical advantage. Towlson et al. [9] observed that relative age, maturation, and anthropometric characteristics contributed to biases in allocation to certain defensive positions among elite youth footballers, with goalkeepers and central defenders generally being taller, heavier, and more advanced in maturation during some developmental stages. More recently, the RAE has also been documented across all playing positions in professional football, although its magnitude may vary by position and competitive context [10]. Nevertheless, most of this evidence has relied on frequency distributions or comparisons across positions, making it difficult to determine whether birth quarter remains associated with playing position after simultaneously accounting for closely related variables such as chronological age, height, and competitive context.

A second issue concerns the potential economic consequences of the RAE. Market value provides an indirect, multifactorial estimate of a footballer's professional valuation and should therefore not be interpreted as a direct measure of performance. Nevertheless, it allows a relevant developmental hypothesis to be examined: earlier access to higher-level training and competition could contribute to differences in professional experience and visibility that remain associated with economic valuation in adulthood. Conversely, relatively younger players who reach senior international football may represent a particularly selected subgroup, potentially attenuating or reversing an early advantage. These competing pathways provide a rationale for examining birth quarter in relation to senior market value without assuming that early-born players will necessarily command higher valuations.

However, the available evidence challenges a linear relationship between early birth within the selection year and higher market value. Doyle and Bottomley [11] found a higher frequency of players born early in the year without observing higher economic valuations or greater playing time. Across U18–U23 age groups, Romann et al. [12] further showed that the relationship between relative age and market value varied with age: the economic advantage initially observed among relatively older players diminished and eventually reversed in age groups approaching senior professional football.

Evidence from professional football reinforces this interpretation. In a sample of 18,429 adult footballers from the top divisions of 54 European countries, Bezuglov et al. [4] observed marked overrepresentation of players born in the first quarter (Q1) compared with the fourth quarter (Q4), but found no statistically significant differences in market value between birth quarters when playing position and competitive level were considered jointly. These findings highlight a relevant paradox: the RAE may remain visible in the composition of elite football without this unequal representation necessarily implying a corresponding professional advantage. However, whether this pattern also occurs in the specific context of the highest level of international competition, where players from different competitive systems are grouped within national teams, remains to be established. Analysing the 48 national teams participating in the 2026 FIFA World Cup allows the associations of birth quarter with playing position and market value to be examined jointly, while accounting for the hierarchical structure arising from players being clustered within national teams.

Previous studies have examined birth-quarter representation in international competitions [13] and its relationship with market value in professional football [4,11,12]. Building on this evidence, the present study examines birth-quarter representation, playing position, and market value within the same multinational cohort from the 2026 FIFA World Cup. Its contribution lies in distinguishing unequal birth-quarter representation from adjusted associations with positional and economic outcomes among players already selected for senior international competition. Multivariable models account for measured individual and contextual characteristics, while the market-value analysis additionally incorporates a national-team random intercept to account for dependence among players representing the same team. This combined approach extends descriptive evidence by assessing the magnitude and precision of birth-quarter associations with both outcomes while accounting for the available covariates and, for market value, national-team clustering.

Against this background, the present study analysed players from the 48 national teams participating in the 2026 FIFA World Cup with two complementary objectives: (i) to determine whether birth quarter was independently associated with playing position after accounting for chronological age, height, and continental confederation; and (ii) to examine whether birth quarter remained independently associated with market value after accounting for the main available individual and contextual factors, while incorporating national team as a clustering level. The study thus aimed to distinguish the persistence of the RAE as unequal representation from its potential associations with tactical and economic characteristics among players who had reached the highest international level. We hypothesised that, despite the overrepresentation of players born early in the year, associations of birth quarter with playing position and market value would be limited after accounting for the main individual and contextual variables. Results were interpreted with attention to the magnitude and precision of the estimates, without treating a lack of statistical significance as sufficient evidence of no association.

2. Materials and Methods

2.1. Study Design

An observational, analytical, cross-sectional study was conducted to evaluate the adjusted associations of calendar birth quarter with positional specialisation and market value among elite international footballers. Descriptive analyses, multivariable regression models, and multilevel linear mixed models were combined with additional procedures to assess the robustness of the main findings.

The analytical framework was structured into a pre-specified hierarchy of primary, sensitivity, and exploratory analyses. Primary analyses evaluated the core study hypotheses: (i) multivariable binary logistic regression models testing the independent association

between birth quarter and allocation into each playing position (one-vs-rest); and (ii) a multilevel linear mixed model (LMM) with a national-team random intercept testing differences in log-transformed market value across birth quarters while accounting for squad-level clustering. Sensitivity analyses evaluated model robustness against alternative specifications, including a multivariable multinomial logistic regression model for playing positions, an ordinary least squares (OLS) linear model for market value, and a multivariable binary logistic regression model predicting membership in the upper quartile (top 25%) of market value. Exploratory analyses examined potential moderation effects via interaction terms (birth quarter × playing position and birth quarter × chronological age) and evaluated indirect pathways via exploratory causal mediation analysis for adult stature.

2.2. Participants

The sample comprised 1248 elite international footballers from the 48 national teams participating in the 2026 FIFA World Cup (26 players per squad). Positional analyses included the full cohort ($N = 1248$). Market-value analyses were conducted using a complete-case approach on players with available Transfermarkt valuations ($n = 827$; 66.3% of the total cohort), nested in 47 national teams. Missing market-value records ($n = 421$; 33.7%) were systematically evaluated against baseline characteristics. Missingness was independent of birth quarter, $\chi^2(3) = 2.73$, $p = 0.434$, providing no evidence that data attrition was associated with the primary exposure. In contrast, missing valuations differed by playing position, $\chi^2(3) = 20.31$, $p < 0.001$ (lowest coverage among goalkeepers, 50.3%), and markedly by continental confederation, $\chi^2(5) = 213.82$, $p < 0.001$, reflecting higher coverage for the Union of European Football Associations (UEFA; 89.4%) and the South American Football Confederation (CONMEBOL; 69.2%) relative to the Confederation of African Football (CAF; 63.5%), the Asian Football Confederation (AFC; 35.0%), and the Oceania Football Confederation (OFC; 53.8%). This structural disparity justified the mandatory inclusion of continental confederation as a fixed covariate and national team as a clustering level in all economic models.

2.3. Data Sources

Official squad lists, players' dates of birth, playing positions, and adult heights were obtained from documentation published by the Fédération Internationale de Football Association (FIFA) for the 2026 World Cup. The database was supplemented with market values from Transfermarkt. Data were extracted on 10 August 2026. Analytically, this valuation date represents a post-tournament measure capturing the consolidated market adjustments implemented following the conclusion of the 2026 FIFA World Cup tournament.

2.4. Variables

The main independent variable was calendar birth quarter, derived from each player's date of birth and classified into four categories: Q1 (January–March), Q2 (April–June), Q3 (July–September), and Q4 (October–December). Q1 served as the reference category in the inferential models. This classification describes birth timing within the calendar year rather than relative age within each player's original eligibility cohort, because federation-specific selection cut-offs during youth development were not reconstructed. A common calendar-year classification was used to provide a consistent descriptive framework across the multinational sample and facilitate comparison with previous football studies [4,11,13]. It does not assume that all players experienced a 1 January eligibility cut-off; therefore, calendar birth quarter is treated as the measured exposure rather than a validated measure of relative age across all developmental systems.

Two main dependent variables were considered. The first was playing position, used as an indicator of positional specialisation. Specific positions were grouped into four categories: goalkeepers (GK), defenders (DF), midfielders (MF), and forwards (FW). For inferential analyses, each position was treated as a binary outcome, membership of that position versus all other positions, in binary logistic regression models. These analyses were complemented by a multinomial logistic regression model, with goalkeeper as the reference category.

The second dependent variable was player market value, expressed in euros. Given its skewed distribution, market value was transformed using the natural logarithm before inferential analyses to reduce skewness and support compliance with model assumptions. Descriptive statistics were presented on the original monetary scale.

Models of playing position were adjusted for chronological age (years), adult height (cm), and continental confederation. Models of market value included the same covariates plus the quadratic term of age (age^2), with playing position additionally incorporated as a fixed factor. The covariate adjustment set was established based on theoretical and empirical criteria governing athletic valuation and tactical sorting. The linear and quadratic terms for age accounted for the well-established concave trajectory of peak performance and financial valuation in professional football. Continental confederation captured macroeconomic disparities across regional football markets. Adult height was incorporated in positional models as a primary biomechanical constraint governing positional sorting (specifically distinguishing goalkeepers and central defenders from other outfield roles). While adult height does not capture biological maturation timing during youth development, exploratory causal mediation analysis provided no evidence that adult height mediated the association between calendar birth quarter and market value (average causal mediation effect (ACME) = 0.0002, 95% confidence interval (CI) [−0.0078, 0.0088], $p = 0.974$; proportion mediated = −0.01%, $p = 0.990$ across 5000 quasi-Bayesian Monte Carlo simulations). All generalised variance inflation factor values (GVIFs) were <1.35 for non-age covariates, indicating no problematic multicollinearity.

2.5. Statistical Analysis

Descriptive statistics were calculated for all study variables. Categorical variables, including birth quarter, playing position, and continental confederation, were summarised using absolute frequencies and percentages. For continuous variables, including age, height, and market value, normality was assessed using the Shapiro–Wilk test and visual inspection of Q–Q plots. Normally distributed variables were expressed as mean ± standard deviation, whereas non-normally distributed variables were summarised using the median and interquartile range. Means and standard deviations were also reported for market value as supplementary descriptive information.

The relationship between birth quarter and positional specialisation was examined in two stages. First, birth quarters were defined according to the calendar year using 1 January as the cut-off, consistent with established international football research [4,11,13]. Pearson's chi-square goodness-of-fit tests compared the observed birth-quarter distributions against a theoretical uniform distribution (25% per quarter; df = 3), both in the overall sample and separately for goalkeepers, defenders, midfielders, and forwards.

Applying an equal quarterly expectation is the standard methodological convention in multinational cohorts comprising 48 federations where standardised, historical census birth data spanning multiple decades are unavailable. Furthermore, prior methodological sensitivity analyses conducted on this international cohort confirmed that while mild seasonal schedules can attenuate bivariate goodness-of-fit significance, population birth seasonality does not account for the pronounced quarterly selection asymmetries observed in elite squads [14]. In addition, an omnibus chi-square test of independence (df = 9)

was performed across the four playing positions, with the effect size quantified using Cramér's V.

Second, to address distinct analytical questions regarding positional allocation, two complementary regression frameworks were implemented. Four separate multivariable binary logistic regression models were specified as the primary approach to evaluate the adjusted odds of being allocated to a specific tactical position compared with the remaining player pool (one-vs-rest specification). This strategy was chosen to yield position-specific, clinically interpretable odds ratios (ORs) without imposing dependency on an arbitrary outfield reference category. All models were adjusted for chronological age, height, and continental confederation. To account for the mutually exclusive and polytomous structure of playing positions, a multivariable multinomial logistic regression model was estimated as a complementary analysis. Goalkeepers were designated as the baseline reference category because of their distinct physiological, anthropometric, and functional demands relative to outfield players, generating relative risk ratios (RRRs) with 95% CIs.

A hierarchical analytical approach was used to examine differences in market value across birth quarters. An LMM included a random intercept for national team to account for clustering and the competitive context shared by players within the same team. A random-intercept structure was chosen to favour parsimony and support numerical convergence and model stability, given the squad size within each national team. An OLS linear regression model was also fitted as a sensitivity analysis and for comparison with conventional approaches in the literature. Model fit was assessed using the Akaike information criterion (AIC), computed from maximum-likelihood fits. Both models included birth quarter (with Q1 as the reference category), age, age squared, height, playing position, and continental confederation as predictors. Model diagnostics included residual plots, assessment of the random-effects distribution and homoscedasticity, and checks for multicollinearity. To facilitate economic interpretation, log-transformed regression coefficients (β) from the multilevel mixed model were converted into relative percentage differences using the standard exponential transformation, $\Delta\% = [\exp(\beta) - 1] \times 100$. To ensure exact mathematical reproducibility, all percentage differences and corresponding 95% confidence intervals were computed directly from exact, unrounded model parameter estimates prior to final tabular rounding.

The multilevel model was further characterised by the residual variance, random-intercept variance, and intraclass correlation coefficient (ICC). The ICC quantified the proportion of the residual variance (i.e., variance not explained by the fixed effects) in log-transformed market value attributable to differences between national teams after adjustment for the included covariates.

Finally, robustness analyses assessed the stability of the main estimates. A multivariable logistic regression model estimated the probability of being in the upper quartile of market value (top 25%). Interaction terms between birth quarter and playing position, and between birth quarter and chronological age, were assessed using Type III F-tests in the OLS specification. An exploratory mediation analysis examined whether adult height mediated the association between early birth within the calendar year (Q1 versus Q2–Q4) and market value. The ACME and average direct effect (ADE) were estimated using 5000 quasi-Bayesian Monte Carlo simulations, adhering to established stability standards for indirect effect sampling distributions. Given the cross-sectional design, these estimates were interpreted as exploratory rather than as evidence of causal mediation. Multicollinearity among predictors was assessed using GVIFs, with values below 1.35 for all non-age covariates. All estimates were reproduced in R version 4.3.3 with *nnet* (version 7.3-19), car (version 3.1-2), *lme4* (version 1.1-35.1), *lmerTest* (version 3.1-3), *performance* (version 0.10.8), and *mediation* (version 4.5.1). A two-sided significance threshold of $p < 0.05$ was adopted for primary

pre-specified hypotheses. Formal family-wise error adjustments were not applied across the multivariable regression models to preserve statistical power for detecting independent associations in observational data. Consequently, non-primary inferential comparisons, specifically those derived from the multinomial logistic regression, were treated as complementary sensitivity analyses and interpreted with caution regarding potential Type I error inflation. Bonferroni-adjusted *p*-values were additionally reported for the three Q4 vs. Q1 contrasts of the multinomial model.

2.6. Ethical Considerations

The study used exclusively publicly accessible secondary sources, specifically official FIFA documentation and Transfermarkt records. No recruitment, intervention, or direct contact with players took place, and no private records were accessed or health-related or other special-category personal data were processed. Given these characteristics, the authors considered that ethics committee review and individual informed consent were not required. No formal exemption was requested.

3. Results

3.1. Birth Quarter and Playing Position

Table 1 distinguishes two analytical questions. The goodness-of-fit tests assess whether birth quarters are equally represented within each playing position, whereas the adjusted logistic models assess whether birth quarter is associated with the odds of occupying a given position rather than any other position, conditional on the included covariates. Goodness-of-fit tests indicated that birth quarters differed significantly from a uniform distribution among goalkeepers, $\chi^2(3) = 12.32$, $p = 0.006$ (Q1/Q4 ratio = 2.47, Cramér's V = 0.17), and defenders, $\chi^2(3) = 8.50$, $p = 0.037$ (Q1/Q4 ratio = 1.50, Cramér's V = 0.08), reflecting small effect sizes. Conversely, birth quarters did not differ statistically from an equal distribution among midfielders, $\chi^2(3) = 6.27$, $p = 0.099$ (Q1/Q4 ratio = 1.32, Cramér's V = 0.08), or forwards, $\chi^2(3) = 1.19$, $p = 0.757$ (Q1/Q4 ratio = 1.14, Cramér's V = 0.04). The omnibus test of independence across positional roles revealed no significant association between birth quarter and playing position across the sample, $\chi^2(9) = 9.56$, $p = 0.387$, indicating a negligible effect size (Cramér's V = 0.051).

Table 1. Birth-quarter distributions by playing position and multivariable logistic regression models.

	Chi-Square Goodness-of-Fit Tests					**Multivariable Logistic Regressions**		
	Q1 ***n*** **(%)**	**Q2** ***n*** **(%)**	**Q3** ***n*** **(%)**	**Q4** ***n*** **(%)**	$\boldsymbol{\chi^2}$ **(*p*) [V]**	**Q2 vs. Q1 [95% CI] (*p*)**	**Q3 vs. Q1 [95% CI] (*p*)**	**Q4 vs. Q1 [95% CI] (*p*)**
GK	47 (32.4%)	42 (29.0%)	37 (25.5%)	19 (13.1%)	12.32 (0.006) [0.17]	0.80 [0.47, 1.34] (0.388)	1.02 [0.60, 1.75] (0.933)	0.53 [0.28, 1.01] (0.055)
DF	126 (29.9%)	103 (24.5%)	108 (25.7%)	84 (20.0%)	8.50 (0.037) [0.08]	0.80 [0.58, 1.10] (0.174)	1.02 [0.74, 1.41] (0.902)	0.94 [0.67, 1.33] (0.724)
MF	99 (26.8%)	107 (29.0%)	88 (23.8%)	75 (20.3%)	6.27 (0.099) [0.08]	1.32 [0.94, 1.86] (0.111)	0.95 [0.67, 1.36] (0.776)	0.99 [0.68, 1.43] (0.945)

Table 1. *Cont.*

	Chi-Square Goodness-of-Fit Tests					Multivariable Logistic Regressions		
	Q1 ***n*** **(%)**	**Q2** ***n*** **(%)**	**Q3** ***n*** **(%)**	**Q4** ***n*** **(%)**	χ^2 **(*p*) [V]**	**Q2 vs. Q1 [95% CI] (*p*)**	**Q3 vs. Q1 [95% CI] (*p*)**	**Q4 vs. Q1 [95% CI] (*p*)**
FW	84 (26.8%)	82 (26.2%)	73 (23.3%)	74 (23.6%)	1.19 (0.757) [0.04]	1.08 [0.76, 1.54] (0.667)	0.98 [0.68, 1.40] (0.896)	1.26 [0.87, 1.83] (0.218)
Total	356 (28.5%)	334 (26.8%)	306 (24.5%)	252 (20.2%)	19.41 (<0.001) [0.07]	—	—	—

Note. $N = 1248$ elite football players. Frequencies are expressed as absolute count n and percentage (%) within each playing position. Unadjusted quarterly distributions were assessed using Pearson's goodness-of-fit chi-square tests against equal quarterly expectations (df = 3): Goalkeepers, $\chi^2(3) = 12.32$, $p = 0.006$; Defenders, $\chi^2(3) = 8.50$, $p = 0.037$; Midfielders, $\chi^2(3) = 6.27$, $p = 0.099$; Forwards, $\chi^2(3) = 1.19$, $p = 0.757$; and Total sample, $\chi^2(3) = 19.41$, $p < 0.001$. The omnibus test of independence across positional roles was $\chi^2(9) = 9.56$, $p = 0.387$ (Cramér's V = 0.051). Odds ratios (OR) and 95% confidence intervals [95% CI] were estimated via multivariable binary logistic regression models for each position (one-vs-rest), using Quarter 1 (Q1) as the reference category and adjusting for age, height, and continental confederation. All market-value analyses are reported in Table 2. Abbreviations: χ^2 = Pearson's Goodness-of-Fit Chi-Square Statistic; CI = Confidence Interval; DF = Defenders; FW = Forwards; GK = Goalkeepers; MF = Midfielders; n = Absolute Subsample Count; N = Total Sample Size; OR = Adjusted Odds Ratio; p = Probability Value; Q1–Q4 = Birth Quarters 1 to 4; V = Cramér's V Effect Size Statistic.

Turning to the adjusted positional associations, interpretation focused on effect magnitudes and the precision of the 95% confidence intervals. The comparisons reported below contrast Q4 with Q1 and concern membership of each position versus all remaining positions, rather than departures from equal birth-quarter representation within a position. In adjusted models, outfield positions showed point estimates close to unity with confidence intervals encompassing the null value (defenders: OR = 0.94, 95% CI [0.67, 1.33], $p = 0.724$; midfielders: OR = 0.99, 95% CI [0.68, 1.43], $p = 0.945$; forwards: OR = 1.26, 95% CI [0.87, 1.83], $p = 0.218$). For goalkeepers, the Q4 versus Q1 comparison indicated a substantial point estimate (OR = 0.53, $p = 0.055$), with a 95% confidence interval [0.28, 1.01] that spans from a 72% reduction in the odds of being a goalkeeper to a trivial 1% increase, reflecting marked uncertainty compatible with both substantially lower odds and negligible differences, given the bounded sample size of goalkeepers ($n = 145$). Complementary multinomial logistic regression yielded findings consistent with this pattern. Using goalkeepers as the reference category, birth quarter was not significantly associated with allocation to defensive (DF vs. GK: Q4 vs. Q1 RRR = 1.74, 95% CI [0.90, 3.37], $p = 0.101$) or midfield positions (MF vs. GK: Q4 vs. Q1 RRR = 1.83, 95% CI [0.90, 3.72], $p = 0.093$). A statistically significant nominal relative risk ratio emerged only for forwards relative to goalkeepers (FW vs. GK: Q4 vs. Q1 RRR = 2.17, 95% CI [1.08, 4.37], unadjusted $p = 0.030$). However, this comparison did not retain statistical significance under formal family-wise adjustment across the three outfield-to-goalkeeper contrasts (Bonferroni-adjusted $p = 0.091$). In the context of the primary analyses and the wider set of statistical evaluations, this multinomial estimate should be interpreted strictly as a complementary finding reflecting the bounded number of late-born goalkeepers ($n = 19$) rather than standalone evidence of preferential offensive sorting.

Table 2. Market value by birth quarter: descriptive statistics, multivariable linear regression, and a multilevel linear mixed model.

	M **(±*SD*)**	***Mdn*** **[IQR]**	**OLS**		**LMM**		
			β (SE)	***p***	**β (SE)**	***p***	**95% CI**
Intercept	—	—	11.13 (1.95)	<0.001	9.297 (1.653)	<0.001	6.06, 12.54
Q1	17.12 (22.53)	8.00 [2.43, 25.00]	—	—	—	—	—
Q2	19.65 (23.75)	10.00 [3.00, 29.00]	0.16 (0.12)	0.172	0.148 (0.097)	0.129	−0.04, 0.34
Q3	18.59 (29.79)	6.00 [2.00, 24.50]	−0.03 (0.12)	0.787	−0.014 (0.100)	0.892	−0.21, 0.18
Q4	18.74 (24.89)	10.00 [3.50, 25.00]	0.05 (0.12)	0.660	0.101 (0.104)	0.332	−0.10, 0.31
Age (years)	—	—	0.35 (0.11)	0.001	0.39 (0.09)	<0.001	0.21, 0.56
Age^2 (years^2)	—	—	−0.008 (0.002)	<0.001	−0.008 (0.002)	<0.001	−0.01, −0.005
Height (cm)	—	—	−0.003 (0.007)	0.681	0.001 (0.006)	0.825	−0.01, 0.01
Playing Position	—	—	$F = 8.61$	<0.001	$F = 17.15$	<0.001	—
Confederation	—	—	$F = 56.06$	<0.001	$F = 9.14$	<0.001	—
τ_{00}	—	—	—	—	0.558	—	—
σ^2	—	—	1.525	—	1.027	—	—
ICC	—	—	—	—	0.352	—	—
Model Fit	—	—	$R^2 = 0.358$ $R^2_{adj} = 0.347$ AIC = 2712.6	<0.001	$R^2_m = 0.344$ $R^2_c = 0.575$ AIC = 2493.7	—	—

Note. $N = 827$ players with valid market value data. Descriptive statistics ($M \pm SD$ and *Mdn* [IQR]) are expressed in millions of Euros (€M). Dependent variable for both OLS and LMM regression models: natural log-transformed market value (log[EUR]). Quarter 1 (Q1) served as the reference category for birth quarters. Playing position was evaluated as a categorical factor with Goalkeepers as reference ($F = 8.61$, $p < 0.001$ in OLS; $F = 17.15$, $p < 0.001$ in LMM). Confederation was evaluated as a categorical factor with AFC as reference ($F = 56.06$, $p < 0.001$ in OLS; $F = 9.14$, $p < 0.001$ in LMM). The random intercept in the LMM accounts for grouping at the National Team level. Birth-quarter coefficients (β) and standard errors (SE) from the LMM are reported to three decimal places. Relative percentage differences [$\Delta\% = (\exp(\beta) - 1) \times 100$] were calculated using unrounded LMM coefficients. Abbreviations: AIC = Akaike Information Criterion; β = Unstandardised Regression Coefficient; CI = Confidence Interval; *F* = Snedecor's F Statistic; ICC = Intraclass Correlation Coefficient (proportion of the variance not explained by the fixed effects that is attributable to National Team grouping); IQR = Interquartile Range; LMM = Multilevel Linear Mixed Model; M = Mean; Md*N* = Median; OLS = Ordinary Least Squares; *p* = Statistically Assessed Probability Value; Q1–Q4 = Birth Quarters 1 to 4; *SD* = Standard Deviation; *SE* = Standard Error; σ^2 = Residual Variance; τ_{00} = Random Intercept Variance.

Figure 1 complements Table 1 by displaying the adjusted probabilities of each playing position across birth quarters and the adjusted odds ratios with their 95% confidence intervals. The goalkeeper model yielded lower estimated odds for Q4 than for Q1, although the confidence interval included 1. For defenders, midfielders, and forwards, the estimated odds ratios were generally closer to 1, with all confidence intervals encompassing the null value.

Overall, the primary adjusted comparisons did not identify statistically significant associations between birth quarter and playing position. These findings do not negate the unequal birth-quarter distributions observed within some positions, as the descriptive and

adjusted analyses address different questions. The precision of the estimates, particularly for goalkeepers, also limits conclusions about the absence of potentially relevant associations.

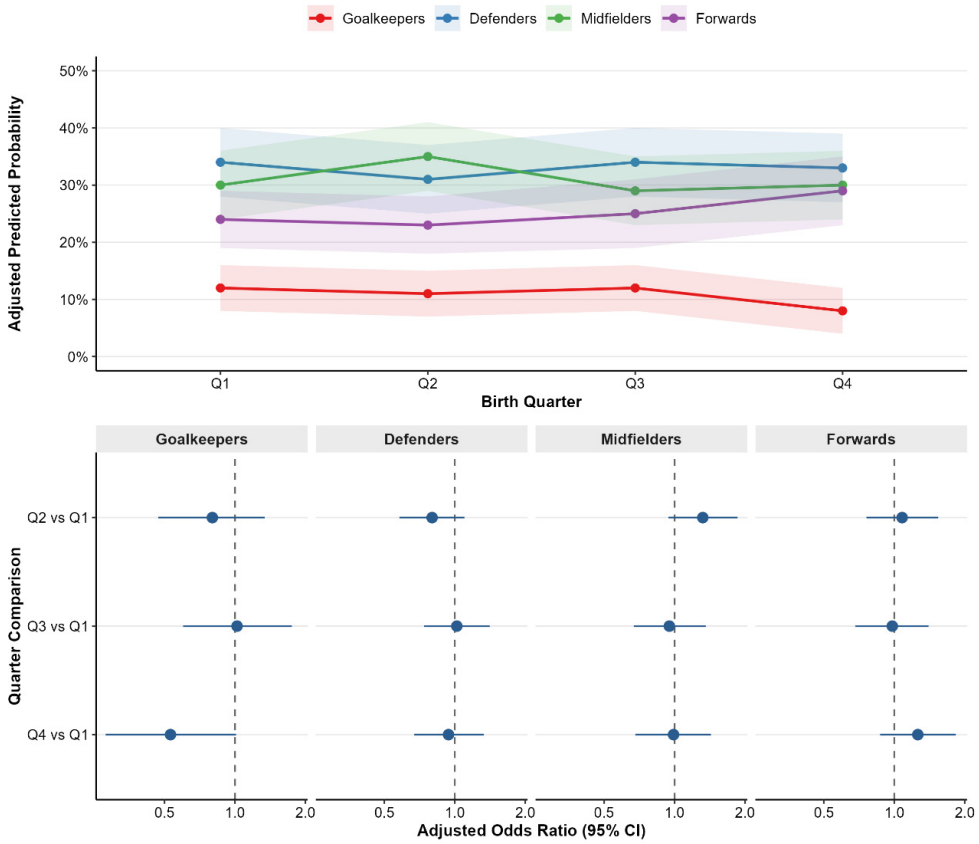


Figure 1. Adjusted probabilities of playing position by birth quarter and adjusted odds ratios with 95% confidence intervals. *Note*. $N = 1248$. The upper panel displays model-adjusted probabilities of playing position across calendar birth quarters. The lower panel displays adjusted odds ratios and 95% confidence intervals for Q2, Q3, and Q4 relative to Q1, estimated from separate binary logistic models comparing each position with all remaining positions. Models were adjusted for chronological age, height, and continental confederation. The dashed vertical lines indicate OR = 1. Q1 = January–March; Q2 = April–June; Q3 = July–September; Q4 = October–December.

3.2. Birth Quarter and Market Value

To examine whether birth quarter was associated with market value among elite footballers, log-transformed market values were analysed using OLS and LMM approaches. Both models were adjusted for age, age^2, height, playing position, and continental confederation. The multilevel model additionally included a random intercept for national team to account for the hierarchical structure of the data. Table 2 presents the descriptive statistics for market value by birth quarter and the results of the OLS and LMM analyses.

Descriptive statistics showed variation in market value across birth quarters. Median values ranged from €6.00 million for players born in Q3 to €10.00 million for those born in Q2 and Q4. Standard deviations were large across all groups, reflecting substantial variability in player market values.

In the multilevel linear mixed model, effect estimates converted to relative percentage differences $[\Delta\% = (\exp(\beta) - 1) \times 100]$ from unrounded model parameters revealed adjusted differences relative to Q1 of +15.9% for Q2 ($\beta = 0.148$, standard error (SE) = 0.097, 95% CI: −4.2% to +40.3%, $p = 0.129$), −1.4% for Q3 ($\beta = -0.014$, SE = 0.100, 95% CI: −19.0% to +20.1%, $p = 0.892$), and +10.7% for Q4 ($\beta = 0.101$, SE = 0.104, 95% CI: −9.8% to +35.8%, $p = 0.332$). While these comparisons did not achieve statistical significance at $p < 0.05$, their

wide confidence intervals encompass both negligible effects and potentially meaningful economic premiums (up to +40.3% for Q2 and +35.8% for Q4). Consequently, these findings should not be interpreted as conclusive evidence of economic parity across quarters, but rather as reflecting substantial estimate dispersion within an internationally clustered cohort. Chronological age, its quadratic term, playing position, and continental confederation were statistically significantly associated with market value in the adjusted models.

Comparison of model fit criteria demonstrated the clear statistical advantage of the multilevel specification over the standard linear approach. The ordinary least squares model yielded an AIC of 2712.6 ($R^2 = 0.358$, R^2adj = 0.347), whereas the multilevel linear mixed model yielded a markedly lower AIC of 2493.7 when refitted by maximum likelihood ($\Delta AIC = -218.8$), indicating better fit after penalising for model complexity. The multilevel model yielded an adjusted ICC of 0.352, indicating that between-team clustering accounted for 35.2% of the variance in log-transformed market value not explained by the fixed effects, corresponding to 23.1% of the total variance (conditional $R^2 = 0.575$). This substantial clustering confirms the statistical necessity of accounting for national-team clustering in elite international football cohorts.

Figure 2 complements the preceding results by showing the distributions of log-transformed market value across birth quarters alongside the estimated national-team random intercepts from the multilevel model. The birth-quarter distributions overlap substantially, with no clear pattern of higher market values among players born in Q1. The random-intercept estimates illustrate the between-team variation remaining after adjustment for the included covariates.

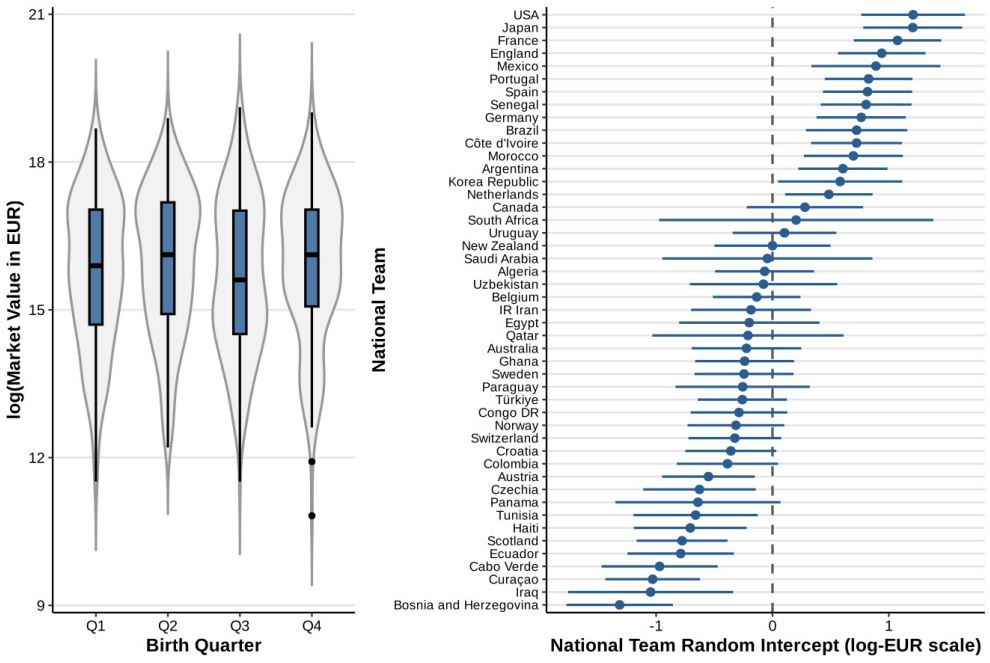


Figure 2. Distribution of log-transformed market value by birth quarter and estimated national-team random intercepts. Note. $N = 827$ players. Left panel: violin plots show the distributions of natural log-transformed market values expressed in euros, with embedded boxplots indicating the median and interquartile range. Birth quarters are defined as Q1 (January–March), Q2 (April–June), Q3 (July–September), and Q4 (October–December). Right panel: points represent estimated national-team random intercepts from the linear mixed model adjusted for age, age squared, height, playing position, and confederation; horizontal bars indicate 95% confidence intervals. The dashed vertical line marks zero.

Overall, the adjusted models did not identify statistically significant associations between birth quarter and market value in this sample of elite international footballers. However, the confidence intervals include potentially relevant differences, so these findings should not be interpreted as conclusive evidence of no association.

Pre-specified exploratory moderation analyses revealed no statistically significant interaction effects on log-transformed market value. Specifically, the interaction between calendar birth quarter and playing position was not statistically significant, $F(9, 803) = 1.37$, $p = 0.195$ (Sum Sq = 18.78), indicating that the relationship between birth timing and economic valuation did not vary across tactical roles. Similarly, the interaction between calendar birth quarter and chronological age was not statistically significant, $F(3, 809) = 0.85$, $p = 0.465$ (Sum Sq = 3.91), indicating that economic valuation across birth quarters remained broadly stable across the age spectrum in this senior cohort.

4. Discussion

The present study examined whether birth quarter was independently associated with playing position and market value among players participating in the 2026 FIFA World Cup after accounting for individual and contextual characteristics. Although the birth-quarter distribution showed an overrepresentation of players born early in the year, the primary adjusted models did not identify statistically significant associations with either outcome. These findings highlight the distinction between unequal representation and the positional and economic associations estimated among players who have reached elite international football. However, the confidence intervals do not conclusively rule out potentially relevant associations.

The persistence of the RAE in professional football and senior international competitions is well documented. In an analysis of 20,401 footballers participating in 47 international tournaments, Pedersen et al. [13] observed that representation bias decreased with age but remained detectable in senior men's competitions. Similarly, Bezuglov et al. [4], using a sample of 18,429 professional players from 54 European countries, identified a considerably higher proportion of players born in Q1 than in Q4. However, the overrepresentation of relatively older players among those reaching the elite does not necessarily imply that birth quarter remains associated with their subsequent professional characteristics. Our results distinguish unequal birth-quarter distributions from associations with playing position and market value within an already selected sample of international footballers. Because only World Cup players were included, the study cannot establish whether these associations differed during youth development or weakened over the course of players' careers.

Research on the transition from youth to senior football helps interpret this apparent paradox. Morganti et al. [6] observed that, despite the initial overrepresentation of relatively older players in Italian youth national teams, relatively younger players who entered the system could exhibit a higher probability of subsequent transition. Likewise, Brustio et al. [5], analysing 9527 players from five major European national-team systems, found marked overrepresentation of players born early in the year at youth levels, alongside higher U17-to-senior and U19-to-senior transition rates for certain relatively younger groups. These findings do not imply an inherent advantage of being born later in the selection year, but challenge a linear interpretation whereby an initial representation advantage necessarily translates into better adult outcomes. Relatively younger players who progress through selection processes may constitute a highly selected group, although the present cross-sectional design cannot directly examine this mechanism.

The first specific objective concerned the association between birth quarter and playing position. Descriptively, goalkeepers and defenders showed greater representation of Q1 players, whereas birth-quarter distributions were more balanced among midfielders and

forwards. One possible interpretation, informed by the developmental literature, is that early positional selection may favour anthropometric and maturational characteristics that differ among players within the same eligibility cohort. However, youth maturation and positional histories were not measured in this study, so the observed senior distribution cannot establish this pathway. In 465 elite U13–U18 footballers, Towlson et al. [9] showed that goalkeepers and central defenders tended to be taller, heavier, and, at some ages, more advanced in maturation than players in other positions. They concluded that relative age, maturation, and anthropometry could bias early allocation towards certain defensive roles.

In the present study, none of the birth-quarter comparisons reached statistical significance in the positional models adjusted for age, height, and confederation. Nevertheless, the goalkeeper estimate for Q4 versus Q1 remained compatible with a potentially substantial association (OR = 0.53; 95% CI [0.28, 1.01]; $p = 0.055$). These findings introduce an important distinction from studies based exclusively on birthdate distributions: a higher proportion of Q1 players within a position does not necessarily establish an independent association between birth quarter and the probability of occupying that position. The adjusted estimates should therefore be interpreted alongside their confidence intervals: the goalkeeper comparison remains compatible with both substantially lower odds of positional membership and little or no difference, while the outfield estimates also retain uncertainty and do not establish the absence of potentially relevant associations. Previous positional findings should be compared at two distinct levels: birth-quarter representation within positions and adjusted associations with positional membership. Gutiérrez Díaz del Campo et al. [15] found that playing position did not significantly influence the extent of the RAE in Spanish youth football, whereas Işık et al. [10] reported unequal birth-quarter distributions across all positional groups in European professional leagues, with the largest departure among goalkeepers. Our descriptive findings partly align with the latter pattern, although departures from uniformity were identified only among goalkeepers and defenders. Differences in age group, competitive context, sample size, and selection into national squads may contribute to this variation, but their contribution cannot be established from the present data. Moreover, neither within-position representation bias nor differences in its magnitude are equivalent to an adjusted association between birth quarter and positional membership.

Accordingly, our findings do not imply that relative age is irrelevant to early positional allocation. Height in this study was measured in adulthood and cannot reconstruct the transient morphological or biological maturation advantages experienced during youth developmental stages. Adult stature may be relevant to positional demands, particularly among goalkeepers; however, the exploratory mediation analysis provided no evidence that height mediated the association between birth quarter and market value. The possibility that relative age and biological maturation are more consequential for positional allocation during early talent identification remains a developmental hypothesis informed by previous research, rather than a finding established by this senior sample. Testing it requires longitudinal studies that record eligibility cut-offs, maturation, and positional trajectories from youth development onwards, including players who do not progress to elite football.

The second specific objective concerned the association between birth quarter and market value. Neither the conventional linear model nor the multilevel model identified statistically significant differences between Q1 and the other birth quarters after adjustment. However, the wide confidence intervals in our study included potentially relevant economic differences (such as an adjusted increase of up to +40.3% for Q2 and +35.8% for Q4), so agreement between the two models should not be interpreted as definitive proof of economic equivalence across birth quarters. The data therefore leave uncertainty about the magnitude and direction of birth-quarter differences in market value within this selected World Cup cohort. The substantial national-team clustering describes an additional feature of the data

structure, but does not resolve that uncertainty or establish the economic irrelevance of birth quarter.

Pérez-González et al. [16] also found no evidence that European professional footballers born early in the year were systematically valued more highly. Romann et al. [12] described an age-dependent pattern among 11,738 U18–U23 players: although Q1 players had higher market values than Q4 players at U19 level, this relationship reversed at U21, U22, and U23 levels, where relatively younger players had higher values. These findings suggest that the relationship between relative age and professional valuation is not static and may change with age as temporary developmental advantages become less influential.

The present economic findings are broadly consistent with Bezuglov et al. [4], who reported no statistically significant market-value differences between birth quarters among adult professionals when playing position and competitive level were considered. However, their European club sample and the present multinational World Cup cohort represent different selection contexts. The age-specific patterns reported by Romann et al. [12] also caution against treating findings from younger age groups as directly interchangeable with those from selected senior internationals. Our study extends this comparison by estimating adjusted associations while accounting for national-team clustering. Nevertheless, differences in samples and analytical approaches prevent direct equivalence between studies, and the uncertainty of our estimates precludes concluding that economic differences are absent.

Taken together, this evidence reinforces the need to distinguish selection, representation, and professional valuation. RAE-related selection advantages may favour the presence of certain players within development systems without ensuring that relatively older players subsequently achieve higher economic valuations. Selection processes could also alter the composition of successive cohorts and the associations observed among those who remain. However, this is an interpretative hypothesis that cannot be tested using the available cross-sectional data. Our findings are restricted to players selected for this competition, among whom no statistically significant birth-quarter differences in market value were detected after adjustment for the included covariates.

An additional finding was the extent of national-team clustering in the multilevel model. The ICC of 0.352 indicated that between-team differences accounted for approximately 35.2% of the variance in log-transformed market value not explained by the covariates (approximately 23.1% of the total variance). This does not imply that national-team membership itself causes higher or lower market value but indicates substantial dependence among players representing the same team. Such clustering may reflect shared factors, including the country's football context, players' clubs and leagues, competitive exposure, or squad composition. These factors were not directly modelled and therefore cannot be identified as explanatory mechanisms from this analysis. The observed clustering supports a multilevel approach that accounts for dependence within national teams.

The sensitivity analyses were consistent with the main models. Birth quarter was not statistically significantly associated with membership of the upper market-value quartile, and the additional analyses did not identify statistically significant interactions with age or playing position or evidence of mediation through height. These analyses should be regarded as supplementary assessments of stability rather than demonstrations that interactions or indirect mechanisms are absent. In particular, adult height cannot reconstruct adolescent maturation, and the cross-sectional design limits any causal interpretation of the exploratory mediation analysis.

Overall, the findings show unequal birth-quarter representation among World Cup players without statistically significant adjusted associations with playing position or market value. Their interpretation must account for estimate precision and the selected nature of the sample; potentially relevant associations and inequalities during youth development

remain possible. The hypothesis that relative age is more consequential during earlier identification, selection, and progression processes is compatible with the literature but was not tested in this study. Evaluating it requires longitudinal research following cohorts from youth development into professional football, including players who do not reach the elite.

4.1. Study Limitations and Future Perspectives

Five main limitations should be considered when interpreting the findings: the cross-sectional design and selection into an already elite World Cup cohort; the use of calendar birth quarter without reconstructing developmental eligibility cut-offs; the nature and incomplete coverage of market-value estimates; residual confounding and the grouping of playing positions; and multiple statistical comparisons.

First, the cross-sectional design allows associations to be estimated but cannot establish causal relationships or reconstruct selection processes, positional specialisation, or changes in economic valuation from youth development onwards. Furthermore, inclusion was conditional on selection for the World Cup. The estimated associations therefore concern players who had progressed through earlier selection processes, which may alter the relationships observed. The findings cannot assess the trajectories of excluded players or rule out relative-age inequalities during sporting progression.

Second, birth quarters were defined according to the calendar year without reconstructing the selection cut-offs used by federations during each player's development. These cut-offs may differ between countries and have changed over time, meaning that calendar birth quarter does not necessarily reflect relative age within the original selection cohort. This potential misclassification should be considered when interpreting the associations. Birth-quarter distributions were also evaluated against a theoretical uniform distribution of 25% per quarter. Although this represents the established baseline in multinational tournament designs, it does not account for country-specific birth seasonality in the general population, which may introduce minor baseline variations [15]. Future investigations focusing on single-nation cohorts should benchmark athletic rosters against exact national census birth records.

Third, Transfermarkt market value is an estimate of economic valuation rather than an observed transfer fee, a market-clearing price, or a direct measure of sporting performance. It may contain measurement error whose magnitude varies by league, competitive context, or player media visibility. Furthermore, the economic analyses included 827 players (66.3% of the total cohort) using a complete-case specification due to unavailable valuations for 421 players (33.7%). While missingness was structurally associated with continental confederation (lower tracking outside Europe and South America), baseline comparisons showed no evidence that it was associated with birth quarter, although coverage was lower among goalkeepers. Nonetheless, this missing-data pattern limits the generalisability of the economic findings primarily to international players integrated into globally monitored football markets.

Fourth, despite adjustment for several individual and contextual characteristics, residual confounding remains possible because other variables were not considered, including contract duration, club and league membership, injury history, playing time, individual performance, and youth national-team experience. In particular, remaining contract duration was not recorded, preventing assessment of whether contractual circumstances contributed to valuation differences or altered the estimated birth-quarter associations. Its omission represents a potential source of residual confounding, although neither the magnitude nor the direction of any resulting bias can be determined from these data. Contract-related effects on negotiated transfer fees should also be distinguished from their possible relationship with Transfermarkt market-value estimates. Future studies should record remaining

contract duration at the valuation reference date and assess whether its inclusion changes the birth-quarter estimates. The national-team random intercept accounts for clustering at that level but does not explicitly model clustering by club or league. Adult height also cannot reconstruct growth and maturation during youth development, while grouping positions into goalkeepers, defenders, midfielders, and forwards may conceal associations with more specific playing roles.

Finally, while multiple statistical procedures, including binary and multinomial logistic models, OLS, LMM, upper-quartile analyses, interaction terms, and mediation models, were undertaken, these were explicitly structured into primary, sensitivity, and exploratory frameworks to control the risk of Type I error. Formal multiplicity corrections were not implemented across all regression parameters. Individual nominal findings from sensitivity procedures, such as the unadjusted association for forwards versus goalkeepers in the multinomial model (unadjusted $p = 0.030$; Bonferroni-adjusted $p = 0.091$), must therefore be interpreted cautiously as complementary observations rather than confirmatory evidence. A lack of statistical significance in primary models does not establish the absence of associations; wide confidence intervals (particularly for goalkeepers and market value comparisons) encompass potentially relevant effect sizes. Consequently, findings must be interpreted in relation to estimate precision and the selected nature of senior World Cup squads.

To address the selected sample and cross-sectional design, future research should follow youth cohorts longitudinally, including players who do not reach senior international football, and record selection, positional trajectories, and changes in economic valuation. Reconstructing federation- and period-specific eligibility cut-offs would allow relative age to be defined within the developmental cohorts actually experienced by each player.

For economic analyses, future studies should seek more complete valuation coverage across football contexts and explicitly assess the consequences of unavailable valuations. Incorporating club and league information would also allow their contribution to player valuation and clustering to be examined alongside national-team membership.

4.2. Practical Applications

For practitioners evaluating senior international players, these findings caution against using calendar birth quarter alone to infer positional suitability or economic advantage. Unequal birth-quarter representation within a position should be distinguished from adjusted associations with positional membership, and non-significant estimates should not be interpreted as proof of equivalent outcomes. Economic comparisons should also consider the national-team clustering observed in this cohort and the incomplete coverage of market valuations. The study does not establish a basis for changing youth selection or positional-development practices; such applications require longitudinal evidence on eligibility cut-offs, maturation, and player progression.

5. Conclusions

This study contributes a combined assessment of calendar birth quarter, playing position, and market value within the same multinational senior World Cup cohort, distinguishing unequal birth-quarter representation from adjusted positional and economic associations. The primary adjusted models did not detect statistically significant birth-quarter associations, but the confidence intervals, particularly for goalkeepers and market value, remained compatible with potentially relevant differences. National-team clustering in log-transformed market value (ICC = 0.352) further supports accounting for dependence within squads when analysing this economic outcome.

These findings are restricted to the highly selected senior players included in the 2026 FIFA World Cup cohort and, for economic analyses, those with available market valuations. They cannot be generalised to youth development or talent-identification pathways. Whether early selection, maturation, or accumulated developmental opportunities contribute to the patterns observed in senior football remains a hypothesis requiring prospective longitudinal research that includes players who do not reach the elite level.

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Data Availability Statement: The data presented in this study are available on request from the corresponding author. The source code used for data acquisition is publicly available through the SportsLabResearch FIFA World Cup 2026 Analytics project. The GitHub repository is available at https://github.com/SportsLabResearch/SportsLabResearch-FIFA-WorldCup-2026-Analytics (accessed on 18 September 2026), and the associated documentation, including information on data sources, processing, validation, and reproducibility, is available at https://sportslabresearch.github.io/SportsLabResearch-FIFA-WorldCup-2026-Analytics/ (accessed on 18 September 2026). An archived version of the project is available through Zenodo at https://zenodo.org/records/21853522 (accessed on 18 September 2026).

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

RAE	Relative age effect
FIFA	Fédération Internationale de Football Association
AFC	Asian Football Confederation
Q1	Birth quarter 1 (January–March)
Q2	Birth quarter 2 (April–June)
Q3	Birth quarter 3 (July–September)
Q4	Birth quarter 4 (October–December)
GK	Goalkeepers
DF	Defenders
MF	Midfielders
FW	Forwards
OR	Odds ratio
RRR	Relative risk ratio
CI	Confidence interval
SE	Standard error
SD	Standard deviation

IQR	Interquartile range
OLS	Ordinary least squares
LMM	Linear mixed model
AIC	Akaike information criterion
ICC	Intraclass correlation coefficient
ACME	Average causal mediation effect
ADE	Average direct effect
UEFA	Union of European Football Associations
CONMEBOL	South American Football Confederation
CONCACAF	Confederation of North, Central America and Caribbean Association Football
CAF	Confederation of African Football
OFC	Oceania Football Confederation
GVIF	Generalised variance inflation factor

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